

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 01)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain the meaning of each of the following terms and explain why such species are given conservation priority.

[8 marks]

EDGE species

.....

.....

.....

Endemic species

.....

.....

.....

Keystone species

.....

.....

.....

Flagship species

.....

.....

.....

0 1 . 2

Explain why the degree of population dispersal is considered when setting priorities.

[2 marks]

.....

.....

.....

.....

10

02 . 1

Explain what a carbon footprint should include when calculated for a car.

[3 marks]

.....

.....

.....

.....

.....

.....

02 . 2

Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

.....

.....

.....

.....

02 . 3

Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

10

03 . 1

Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis

☐

B Radio tracking

☐

C Lincoln index (mark-release-recapture)

☐

D Leg rings

☐

03 . 2

Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

.....

.....

.....

.....

03 . 3

Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

.....

.....

.....

.....

5

04

Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

04 . 1

Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer

%

0 4 . 2

Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

0 4 . 3

Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

0 4 . 4

Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

Do not write outside the box

Extra space

15

0 5 . 1

Describe **two** case studies of biological control, one successful and one that caused problems.

[4 marks]

0 5 . 2

Explain how pheromone traps can be used in pest control.

[2 marks]

0 5 . 3

Explain why the use of antibiotics as growth promoters in livestock is a concern for human health.

[3 marks]

0 5 . 4

Give **one** way farmers can support crop pollinators.

[1 mark]

10

0 6 . 1

Explain why the potential to increase food supplies by cultivating new land is limited.

[2 marks]

0 6 . 2

Explain why importing food may reduce the availability of food in the area where it is produced.

[2 marks]

0 6 . 3

Explain why organic agriculture is more than simply 'not using pesticides and artificial fertilisers'.

[2 marks]

Evaluate the strategies that could be used to make agriculture more sustainable.

Do not write outside the box

Extra space

15

07 . 1

Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

- A Thermal pollution of a river by a power station
- B Flooding a valley to create a reservoir
- C Lowering the water table by groundwater abstraction
- D Acid mine drainage entering a stream

☐
☐
☐
☐

07 . 2

Give **four** human activities that cause complete habitat destruction.

[4 marks]

1

2

3

4

5

0 8 . 1

Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

1

2

0 8 . 2

Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

0 8 . 3

Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

5

09 . 1

Give **two** natural processes that are driven by solar energy.

[2 marks]

1

.....

2

.....

09 . 2

Explain why natural processes can take place at low temperatures.

[2 marks]

.....

.....

.....

.....

09 . 3

Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

.....

.....

5

10 . 1

Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

Do not write outside the box

1	0	.	2
---	---	---	---

Explain why growing coral reefs help to resist climate change.

[2 marks]

1	0	.	3
---	---	---	---

Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Extra space

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

15

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
---	---	---	---

Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

OR

1 1 . 2

Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

END OF QUESTIONS